NANYANG POLYTECHNIC

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Our Ref: ITDFPA (220429P) Date: 25 Jan 25

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Javerine Tan Jing Xuan, T****853E, is a student at the School of Information Technology at Nanyang Polytechnic. She enrolled in April 2022 and is expected to graduate in May 2025 with a Diploma in Applied AI & Analytics. Throughout her studies, Javerine has achieved 4 Distinctions and 15 As over 4 semesters.

Javerine is an avid learner who has demonstrated the ability to acquire new knowledge and skills beyond the standard curriculum. Despite rigorous academic commitments, she successfully obtained industry-recognized certifications such as Microsoft Power BI Data Analyst Associate, AI Singapore AI4I Foundations in AI, and Prompt Design in Vertex AI by Google, further solidifying her technical competence and readiness for industry demands.

During her 2nd year, Javerine participated in an overseas exchange program at Gachon University in Korea, where she studied for one semester. During this time, she excelled in modules such as Machine Learning, Web Database Programming, Principles of Accounting, Consumer Behavior Theory, and Business English - Presentation. Her outstanding performance earned her straight A's and a perfect GPA of 4.5/4.5, further showcasing her adaptability, dedication, and academic excellence in a global learning environment. Her project was even well praised by the professors and shared with future batches of students.

Outside academics, Javerine has actively participated in various competitions, where she and her team achieved 1st Runner up among competing teams from multiple institutes of higher learning during the Build on Singapore Competition. Their success brought pride to Nanyang Polytechnic, further emphasizing Javerine's collaborative and competitive spirit.

In her final year of study, Javerine interned at Home Team Science and Technology (HTX) for 24 weeks, as part of the Chief Data Officer department. During her internship, she undertook key responsibilities, including developing a platform to track data requests from various agencies and ensuring the timely purging of data, designing database diagrams and deploying databases using Azure, as well as independently learning and utilizing tools like Flask, Azure Documentation AI, and RetrievalAugmented Generation (RAG) models. Javerine also collaborated with a teammate to develop a RAGbased chatbot, enabling users to query and retrieve information about PDPA policies and data masking practices. The chatbot decomposed user queries, performed AI-based searches within the organization's documentation, and returned concise and refined answers. Its conversational capabilities allowed it to retain context, providing a seamless user experience. Her exceptional ability to adapt, learn new technologies independently, and deliver impactful solutions demonstrated her strong potential and left a lasting positive impression on her team.

For her Year 3 semestral project, Javerine demonstrated exceptional creativity and technical skills in the area of Multiplayer Game Analysis. She conceptualised, designed, and implemented a comprehensive data streaming pipeline capable of continuously collecting and processing data from Reddit. This pipeline ingested raw data, performed cleaning and preprocessing tasks, and seamlessly streamed the refined data into a dynamic dashboard built with Databricks. The dashboard Javerine

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developed provided real-time insights into live data streams. Her work focused on analysing comments made by Reddit users about various multiplayer games, employing advanced topic modelling and sentiment analysis techniques. Through these analyses, she identified public opinions, extracted common discussion topics, and evaluated how these insights could influence the popularity and perception of multiplayer games.

Javerine demonstrated proficiency in data modelling, strong coding capabilities, and effective teamwork to ensure the successful completion of the project. Her innovative approach and practical application of machine learning techniques in data analytics not only demonstrated her technical proficiency but also her ability to translate raw data into actionable business strategies. Her project stands out as an excellent example of applying academic knowledge to solve real-world challenges in the gaming industry. This project underscored her ability to apply classroom learning to solve realworld problems.

Javerine is a diligent, responsible, and hardworking individual, consistently rising to meet challenges with confidence and determination. These qualities will undoubtedly support her success in future endeavours.

Ms Teo Miow Ting Course

Manager

School of Information Technology